1 Case Study 5: Detecting Emerging Topics

1.1 Objective

Identify emerging topics in scientometrics research (2015–2023) using keyword burst detection and topic modelling, and validate findings against known developments in the field.

1.2 Setup

```r
library(tidyverse)
library(openalexR)
library(quanteda)
library(quanteda.textstats)
library(topicmodels)
library(tidytext)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

1.3 Data acquisition

```r
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2015-01-01",
  to_publication_date = "2023-12-31",
  type = "article",
  options = list(sample = 600, seed = 42)
)

text_df <- works |>
  filter(!is.na(abstract), nchar(abstract) > 50) |>
  transmute(
    doc_id = id,
    text = paste(display_name, abstract, sep = ". "),
    year = year(publication_date)
  )

cat(glue("Documents: {nrow(text_df)}\n"))
```

```
#> Documents: 109
```

1.4 Keyword burst detection

```r
corp <- corpus(text_df, docid_field = "doc_id", text_field = "text")
docvars(corp, "year") <- text_df$year

toks <- tokens(corp, remove_punct = TRUE, remove_numbers = TRUE) |>
  tokens_tolower() |>
  tokens_remove(stopwords("en")) |>
  tokens_remove(c("study", "paper", "results", "research", "analysis",
                  "also", "however", "using", "based"))

dfmat <- dfm(toks) |> dfm_trim(min_termfreq = 10, min_docfreq = 5)

kw_by_year <- quanteda::convert(dfm_group(dfmat, groups = year), to = "data.frame") |>
  pivot_longer(-doc_id, names_to = "term", values_to = "count") |>
  rename(year = doc_id) |>
  mutate(year = as.integer(year))

growth <- kw_by_year |>
  group_by(term) |>
  filter(sum(count) >= 20) |>
  arrange(year) |>
  mutate(growth = (count - lag(count)) / pmax(lag(count), 1)) |>
  ungroup() |>
  filter(!is.na(growth))

recent_growth <- growth |>
  filter(year >= 2021) |>
  group_by(term) |>
  summarise(mean_growth = mean(growth), total = sum(count), .groups = "drop") |>
  filter(mean_growth > 0) |>
  arrange(desc(mean_growth))
```

```r
recent_growth |>
  head(15) |>
  mutate(term = fct_reorder(term, mean_growth)) |>
  ggplot(aes(x = mean_growth, y = term)) +
  geom_col(fill = palette_sci(1)) +
  scale_x_continuous(labels = scales::percent) +
  labs(x = "Mean growth rate (2021-2023)", y = NULL) +
  theme_sci()
```

1.5 Keyword trajectories

```r
emerging_terms <- recent_growth |> head(6) |> pull(term)

kw_by_year |>
  filter(term %in% emerging_terms) |>
  ggplot(aes(x = year, y = count, colour = term)) +
  geom_line(linewidth = 0.8) +
```

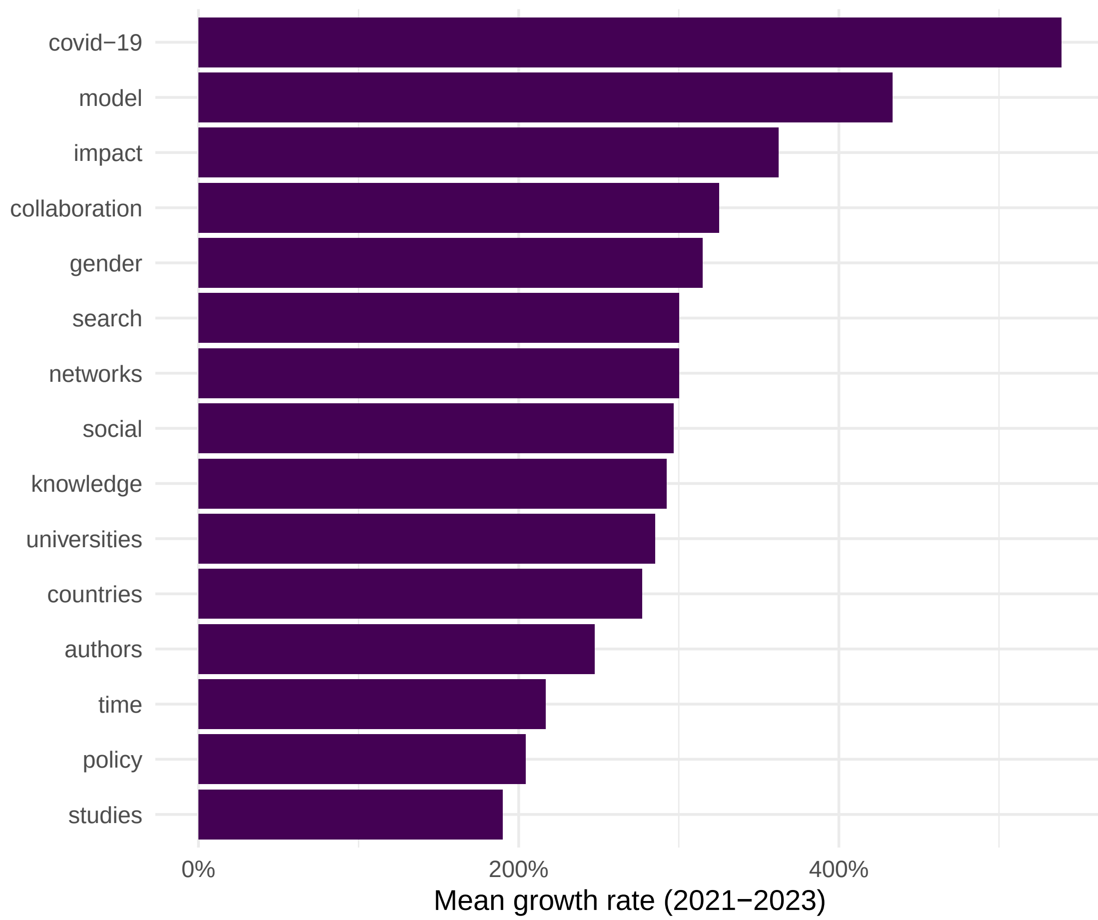

Figure 1: Top 15 fastest-growing keywords (2021–2023).

```r
  geom_point(size = 1.5) +
  scale_colour_manual(values = palette_sci(length(emerging_terms))) +
  labs(x = "Year", y = "Frequency", colour = "Keyword") +
  theme_sci()
```

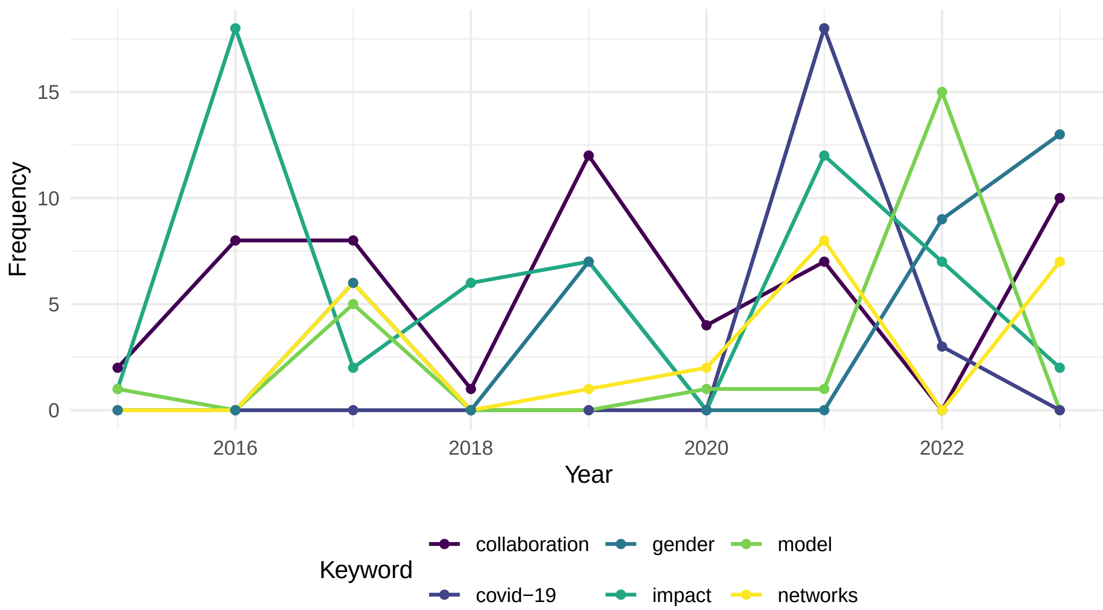

Figure 2: Temporal trajectories of selected emerging keywords.

1.6 Topic modelling

```r
dtm <- quanteda::convert(dfmat, to = "topicmodels")
lda <- LDA(dtm, k = 8, control = list(seed = 42))

lda_topics <- tidy(lda, matrix = "beta") |>
  group_by(topic) |>
  slice_max(beta, n = 8) |>
  ungroup()
```

```r
lda_topics |>
  mutate(term = reorder_within(term, beta, topic)) |>
  ggplot(aes(x = beta, y = term, fill = factor(topic))) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ paste("Topic", topic), scales = "free_y", ncol = 4) +
  scale_y_reordered() +
  scale_fill_manual(values = palette_sci(8)) +
  labs(x = "Word probability", y = NULL) +
  theme_sci(base_size = 9)
```

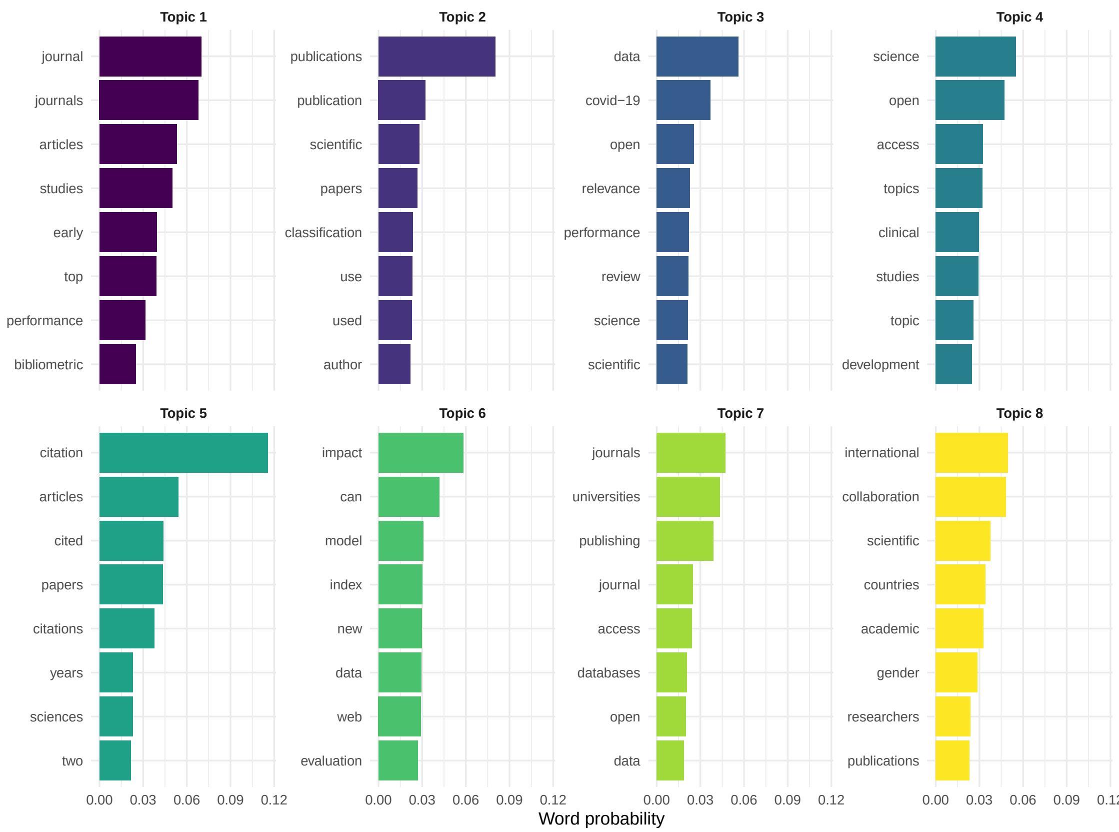

Figure 3: Top terms per LDA topic.

1.7 Topic prevalence by year

```r
doc_topics <- tidy(lda, matrix = "gamma") |>
  left_join(text_df |> transmute(document = doc_id, year), by = "document")

topic_by_year <- doc_topics |>
  group_by(year, topic) |>
  summarise(mean_gamma = mean(gamma), .groups = "drop")

ggplot(topic_by_year, aes(x = year, y = mean_gamma, colour = factor(topic))) +
  geom_line(linewidth = 0.7) +
  scale_colour_manual(values = palette_sci(8)) +
  labs(x = "Year", y = "Mean topic proportion", colour = "Topic") +
  theme_sci()
```

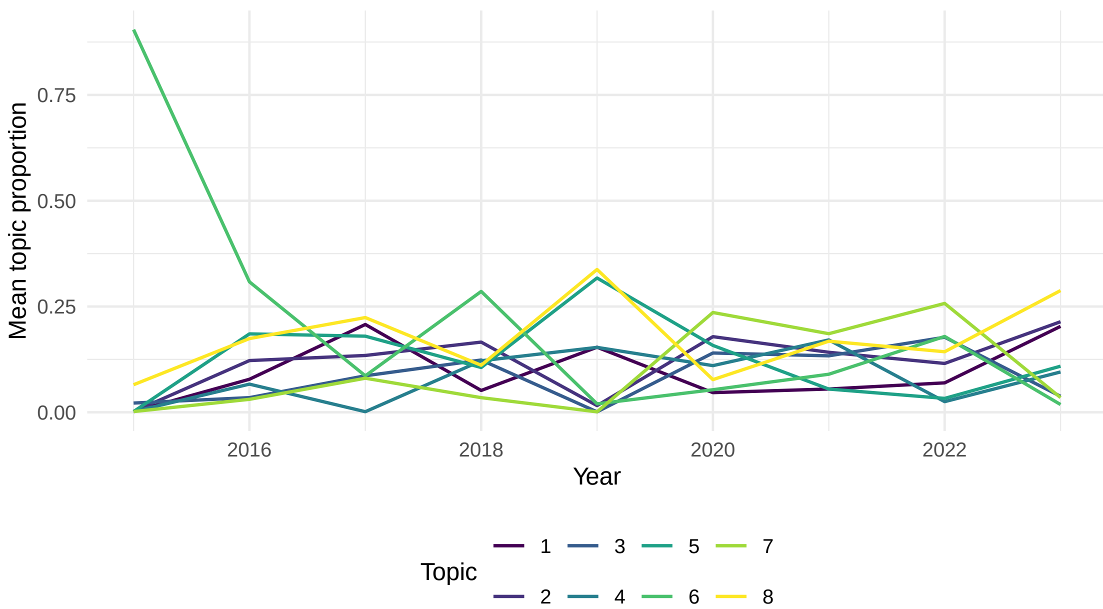

Figure 4: Topic prevalence over time.

1.8 Key findings

1. **Emerging keywords** align with known developments: open science, AI/machine learning applications, equity and diversity, and preprints.
2. **Topic evolution** shows gradual shifts in emphasis rather than abrupt changes, consistent with a mature field incorporating new tools.
3. **Growth rates** should be interpreted cautiously: terms with low baseline frequencies can show high percentage growth from small increases.
4. **Validation**: the topics identified by LDA correspond to recognisable subfields, lending credibility to the unsupervised approach.

1.9 Lessons learned

- Combining keyword growth rates with topic models provides complementary perspectives: growth rates capture individual terms, while topics capture thematic clusters.
- Database coverage growth can create artefactual "emergence." Always normalise by total corpus size.
- Emerging topics in scientometrics mirror broader trends in science policy (open access mandates, equity initiatives, AI adoption) [@kleinberg2003].